

Geographic Apportionment of Sablefish, *Anoplopoma fimbria*, Harvest in the Northeastern Pacific Ocean

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ABSTRACT

An age-structured model of the sablefish population in the northeastern Pacific Ocean that includes movement, survival, and recruitment was constructed to examine yield per recruit and harvest policies. The model includes a Markov transition matrix that specifies the probability that a fish of a given age will move from one fishery regulatory area to another in one year. Estimates of yield per recruit are sensitive to assumptions about the geographic distribution of recruitment. If most recruitment occurs in the eastern areas of the Gulf of Alaska, then fisheries in the western regions will benefit from low exploitation rates. Conversely, if recruitment occurs in all geographic areas, then fisheries in the eastern area will benefit from low exploitation rates. In general, when evaluated under the current annual exploitation rate of about 10% over a 100-year period, policies that apportion harvest among regulatory areas based on areal estimates of available biomass or on the steady-state distribution of biomass computed from the transition matrices gave similar results. The current policy of apportionment is based on a weighted moving average of areal estimates of available biomass. This policy adapts to current information about geographic distribution of biomass, reduces the effects of annual fluctuations in biomass estimates due to measurement error, and does not require estimates of movement probabilities for implementation.

Introduction

The apportionment of catch limits among geographic areas for many groundfish fisheries is based on the geographic distribution of estimated abundance. Surprisingly, few studies have compared alternative methods of apportioning catch limits, partly because of sparse information on the spatial aspects of many fish populations. The purpose of this paper is to explore alternative methods of apportioning the annual catch limits for a highly migratory species: sablefish, *Anoplopoma fimbria*, in the northeastern Pacific Ocean.

Plans are under way to manage the United States fishery for sablefish in the northeastern Pacific Ocean by individual transferable fishing quotas (IFQ's) on an areal basis. Under this management system, an individual will own the right to a certain portion of the catch limit within an area. Thus, apportioning the catch limit among areas can have important social and economic effects.

Annual catch limits for sablefish in waters managed by the North Pacific Fishery Management Council (NPFMC) are typically apportioned among fishery regulatory areas in direct proportion to geographic distribu-

tion of estimated biomass (Gulf of Alaska Groundfish Plan Team¹). Tagging experiments indicate rapid mixing of sablefish among regulatory areas; the pattern of movement is related to fish size and age (Heifetz and Fujioka, 1991). Movement is primarily westward for small, mostly immature sablefish, and eastward for large, mature sablefish. Large sablefish in the Eastern Gulf of Alaska regulatory area (EGOA) tend to remain there. This pattern of movement has not been directly considered in the development of harvest strategies in the past.

In this paper, we use movement rates estimated from tagging data reported by Heifetz and Fujioka (1991) together with other sablefish population parameters to model the movement, survival, growth, and recruitment of an age-structured population. Including movement rates in a population model is an important step towards understanding the spatial dynamics of an exploited population.

Our study is divided into two parts, each with different objectives. First, yield per recruit by area is analyzed to explore the implications of movement and assumptions about the geographic distribution of recruitment with regard to sablefish population dynamics. Second, alternative strategies for geographic distribution of harvest are evaluated. Trends in yield, yield variability, reproductive potential, and catch per unit of effort are examined.

Methods

Model Description

In our population dynamics model, we attempt to accommodate the dynamics of the survival, movement, and recruitment process and the methods used by the NPFMC for setting sablefish catch limits in the northeastern Pacific Ocean (Fig. 1). Typically, a biomass-based acceptable biological catch (ABC) for all regulatory areas combined is computed by multiplying available biomass B for all areas combined by an exploitation rate μ . This ABC is then apportioned to the five regulatory areas based on some policy. We equate ABC with a catch limit and assume that the entire limit is taken each year. Because sablefish in the northeastern Pacific Ocean are a fully utilized species and managed on a quota system, this assumption is approximately correct.

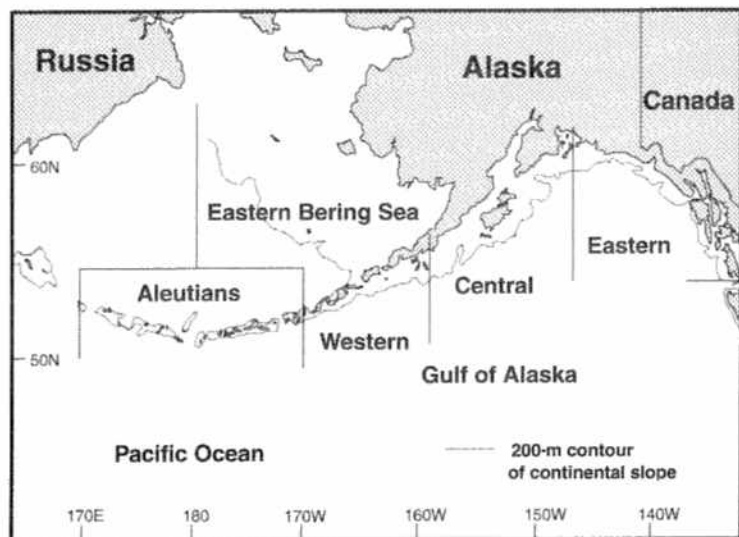


Figure 1
Northeastern Pacific Ocean, showing North Pacific Fishery Management Council regulatory areas.

Given numbers at age a in area r at the beginning of year t , $N(t, a, r)$, weight at age, $w(a)$, availability at age, $s(a)$, and apportionment (proportion of catch limit allotted to area r in year t), $P(t, r)$, available biomass is

$$B(t, a, r) = N(t, a, r)w(a)s(a).$$

The total catch limit among areas is

$$Q(t) = \mu B(t),$$

which is apportioned to areas as

$$Q(t, r) = P(t, r)Q(t).$$

Exploitation rate by area is

$$\mu(t, f) = \frac{Q(t, r)}{\sum_a B(t, a, r)}.$$

Catch in numbers at age is then obtained as

$$C(t, a, r) = \mu(t, r)N(t, a, r)s(a).$$

For convenience, natural mortality M is assumed to occur continuously throughout the year, and fishing is assumed to occur instantaneously at the beginning of a year. Abundance in each area after fishing and natural mortality is

$$N^*(t, a, r) = (N(t, a, r) - C(t, a, r))(\exp(-M)).$$

¹ Gulf of Alaska Groundfish Plan Team. 1992. Current status of stocks and acceptable biological catches. In Stock assessment and fishery evaluation report for the 1993 Gulf of Alaska groundfish fishery, p. 13-33. N. Pac. Fish. Manage. Council, P.O. Box 103136, Anchorage, AK 99510-2252.

Table 1

Estimates of population parameters for sablefish in the northeastern Pacific Ocean. Fraction mature and availability at age are from Sigler and Fujioka (1993). Growth parameters for length (L ; cm) and weight (w ; kg) are adapted from Lowe et al. (1991).

	Age, a									
	3	4	5	6	7	8	9	10	11	12–75
Natural mortality, M	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
Maturity, $m(a)$	0.002	0.032	0.232	0.629	0.868	0.949	0.977	0.988	0.993	1.00
Availability, $s(a)$	0.15	0.32	0.48	0.63	0.73	0.82	0.88	0.93	0.97	1.00
von Bertalanffy length–age model					Allometric weight–length model					
$L_a = L_\infty [1 - \exp(-k(a - t_0))]$					$w(a) = cL_a^b$					
$L_\infty = 80.550$					$c = 2.72 \times 10^{-6}$					
$k = 0.171$					$b = 3.325$					
$t_0 = -3.220$										

Movement is assumed to occur instantaneously at the end of a year. Given the Markov matrix of annual movement rates $\Theta(a)$ where the (i,j) th element of $\Theta(a)$ is the proportion of age- a fish that move from area i to area j in a year, the row vector of numbers at age $a+1$ at the beginning of year $t+1$ after movement is

$$\mathbf{N}(t+1, a+1) = \mathbf{N}^*(t, a)\Theta(a). \quad (1)$$

Quinn et al. (1990a) describe a similar population dynamics model with movement. The order of events in our formulation is different from theirs. We have movement occur at the end of a year, whereas they have movement occur at the beginning of a year. The order of events is somewhat arbitrary, and alternative orders should not produce substantially different results (Getz and Haight, 1989).

Table 1 summarizes population parameters for natural mortality, weight, availability, maturity, and growth for sablefish in the northeastern Pacific Ocean. Although male and female sablefish have different growth patterns, we use growth parameters for both sexes combined because movement rates are estimated for both sexes combined. Movement rates given in Heifetz and Fujioka (1991) are based on fish length. We convert these length-specific movement rates to age-specific movement rates by using von Bertalanffy growth parameters (Table 2).

Geographical Yield per Recruit

First, we investigate the effects on yield of different geographic distributions of recruitment. Assuming that recruitment is independent of spawning biomass allows

Table 2

Annual movement rates from one area to another for different age groups of sablefish; adapted from Heifetz and Fujioka (1991).

From area i	To area j				
	EGOA	CGOA	WGOA	EBS	AL
Age 3–4					
EGOA	0.489	0.378	0.109	0.012	0.012
CGOA	0.194	0.484	0.229	0.047	0.046
WGOA	0.078	0.322	0.308	0.145	0.147
EBS	0.010	0.073	0.160	0.710	0.047
AL	0.005	0.040	0.093	0.049	0.813
Age 5–8					
EGOA	0.712	0.227	0.054	0.003	0.004
CGOA	0.273	0.476	0.199	0.023	0.029
WGOA	0.134	0.409	0.285	0.072	0.100
EBS	0.029	0.165	0.246	0.494	0.066
AL	0.013	0.077	0.128	0.034	0.748
Age ≥ 9					
EGOA	0.751	0.200	0.044	0.002	0.003
CGOA	0.472	0.426	0.074	0.011	0.017
WGOA	0.262	0.116	0.517	0.046	0.059
EBS	0.095	0.181	0.259	0.287	0.178
AL	0.047	0.103	0.155	0.018	0.677

EGOA, Eastern Gulf of Alaska; CGOA, Central Gulf of Alaska; WGOA, Western Gulf of Alaska; EBS, Eastern Bering Sea; AL, Aleutians.

use of the basic-yield-per-recruit model of Beverton and Holt (1957) to compute expected yield. In our analysis, we assume that recruitment, which begins at age 3, is independent of spawning biomass but dependent on

area. That is, recruitment for all areas combined is held constant, but recruitment may differ among areas. Given that recruitment starts at age 3, the abundance of a cohort at age 3 is

$$N(t, 3, r) = R(r).$$

The population dynamics model simulates a 75-year period with $\mu = 0$ to initialize the population; then yield per recruit $Y(r)/R$ is calculated for each area separately over a range of $0.0 \leq \mu \leq 0.7$ as

$$\frac{Y(r)}{R} = \frac{\mu \sum_a s(a)N(a, r)w(a)}{R},$$

where R is recruitment for all areas combined. Because R is a constant on both sides of the above equation, it can be eliminated to compute $Y(r)$. Apportionment among areas is set in proportion to available biomass in each area. Thus the exploitation rate among areas was constant, i.e., $\mu = \mu(r, t)$.

We construct three alternative hypotheses concerning the geographic distribution of annual recruitment to examine effects on areal yields (Table 3). Hypothesis A: recruitment occurs equally in all areas. Hypothesis B: most recruitment occurs in the EGOA, with a progressively lower proportion in the more westward areas. Hypothesis C: all recruitment occurs in the EGOA.

These hypotheses represent a range that we believe reasonably approximates the uncertainties about geographic distribution of recruitment. Considered in development of these hypotheses is the pattern of sablefish movement (Heifetz and Fujioka, 1991). The westward movement of immature sablefish may compensate for the centering of recruitment in the eastern areas. Thus two of the three hypotheses have the high-est proportion of recruitment in the EGOA. Estimation

of the geographical distribution of recruitment and assignment of probabilities to alternative hypotheses may be possible from survey data. This is not within the scope of our study.

Alternative Apportionment Policies

In this part of the study, we modify the simulation model of the yield-per-recruit analysis to add biological realism and measurement error. Stochasticity is incorporated by basing recruitment on past observations (Sigler and Fujioka, 1993). Each year a recruitment level is randomly chosen from the 12 past observations (Fig. 2). This method does not assume any functional form to the stock-recruitment relationship, but instead assumes there is no relationship between stock size and resulting recruitment and that recruitment levels observed in the past are equally likely to occur in the future. Recruitment is distributed among the areas according to the three hypotheses of the yield-per-recruit analysis.

Four alternative policies of apportionment of sablefish catch quotas among areas are evaluated.

Policy 1—Constant apportionment among areas. Because there are five areas, all $P(t, r) = 0.20$.

Policy 2—Set the areal catch limits proportional to estimated available biomass in each area. Given estimated available biomass $\hat{B}(t, r)$ summed over all ages in an area, apportionment is

$$P(t, r) = \frac{\hat{B}(t, r)}{\sum_r \hat{B}(t, r)},$$

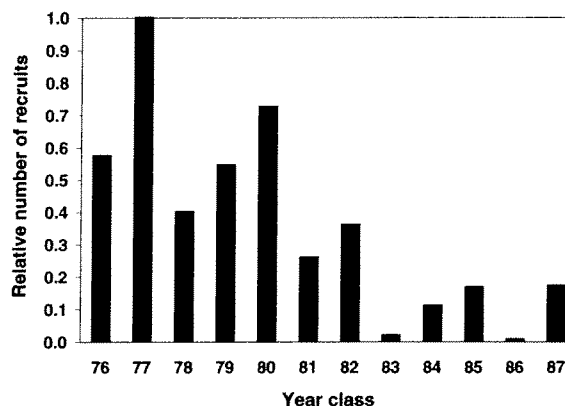


Figure 2

Estimated relative number of sablefish recruits, 1976–87 (Sigler and Fujioka, 1993).

Table 3

Three hypotheses for the geographic distribution of annual recruitment of sablefish.

Hypothesis	Area ¹				
	EGOA	CGOA	WGOA	EBS	AL
A	0.20	0.20	0.20	0.20	0.20
B	0.40	0.30	0.20	0.05	0.05
C	1.00	0.00	0.00	0.00	0.00

¹ EGOA, Eastern Gulf of Alaska; CGOA, Central Gulf of Alaska; WGOA, Western Gulf of Alaska; EBS, Eastern Bering Sea; AL, Aleutians.

where

$$\hat{B}(t, r) = B(t, r) + \varepsilon(t).$$

Variable $\varepsilon(t)$ is a normally distributed random error term with mean 0 and constant coefficient of variation (CV) (i.e., variance = $B^2(t, r)CV^2$). This error term represents measurement error associated with the estimation of available biomass. The CV is estimated to be 0.10 based on analysis of areal variability observed in the annual longline survey of sablefish in the Gulf of Alaska (Sigler²).

Policy 3—Set the catch limits proportional to a weighted moving average of apportionment given in policy 2. The most recent year is given a weight of two, and the prior four years are given a weight of one. This is the present method of apportionment (Gulf of Alaska Groundfish Plan Team¹).

Policy 4—Apportion areal catch limits on the basis of the estimated steady-state distribution of available biomass. Given Equation 1, a steady-state distribution of available biomass exists for each level of exploitation and recruitment hypothesis. As long as the $\Theta(a)$'s do not change, the distribution of available biomass will approach a limit. The expected value of the steady-state distribution is determined by simulation. The population dynamics model is simulated for 200 years with stochastic recruitment. This simulation is replicated 200 times, each with a different random sequence of recruitment. The apportionments for this policy are based on the mean of the 200 replications of the geographic distribution of available biomass at the end of the 200-year period. Exploitation rate among areas is constant, i.e., $\mu = \mu(r, t)$. Table 4 shows the apportionments based on this policy. Stochastic recruitment causes variability in the estimates of the steady-state distribution of available biomass.

Performance Criteria—Two hundred replications over 100 years are performed for each combination of the four apportionment policies and three recruitment hypotheses, giving 12 scenarios. A common random-number sequence is used for each scenario. Exploitation rate μ is set at 0.10, which approximates the current level of exploitation (Fujioka³; Lowe⁴). An ex-

Table 4

Apportionment of sablefish harvest among areas for three hypotheses of the geographic distribution of annual recruitment based on the mean steady-state distribution of available biomass for 200 replicates. Exploitation rate equals 0.10. The coefficient of variation is shown in parentheses.

Hypothesis	Area ¹				
	EGOA	CGOA	WGOA	EBS	AL
A	0.41 (0.07)	0.26 (0.02)	0.15 (0.03)	0.05 (0.19)	0.13 (0.08)
B	0.45 (0.05)	0.28 (0.04)	0.15 (0.03)	0.04 (0.13)	0.09 (0.04)
C	0.52 (0.03)	0.27 (0.04)	0.13 (0.02)	0.02 (0.07)	0.06 (0.07)

¹ EGOA, Eastern Gulf of Alaska; CGOA, Central Gulf of Alaska; WGOA, Western Gulf of Alaska; EBS, Eastern Bering Sea; AL, Aleutians.

ample of the apportionments resulting from a typical replication for recruitment hypothesis B is shown in Figure 3.

To evaluate the effects of different apportionment policies, we compare attributes among policies. These attributes include average annual yield, yield variability, risk, and catch per unit of effort (CPUE). Average annual yield is the mean of the 200 replications. Yield variability is the mean of the 200 replications for the coefficient of variation (CV) of the annual yield. Risk is arbitrarily defined as the proportion of the time that spawning biomass in an area was less than 30% of the unfished level. Spawning biomass in an area is

$$SB(t, r) = \sum_a N(t, a, r) m(a) w(a).$$

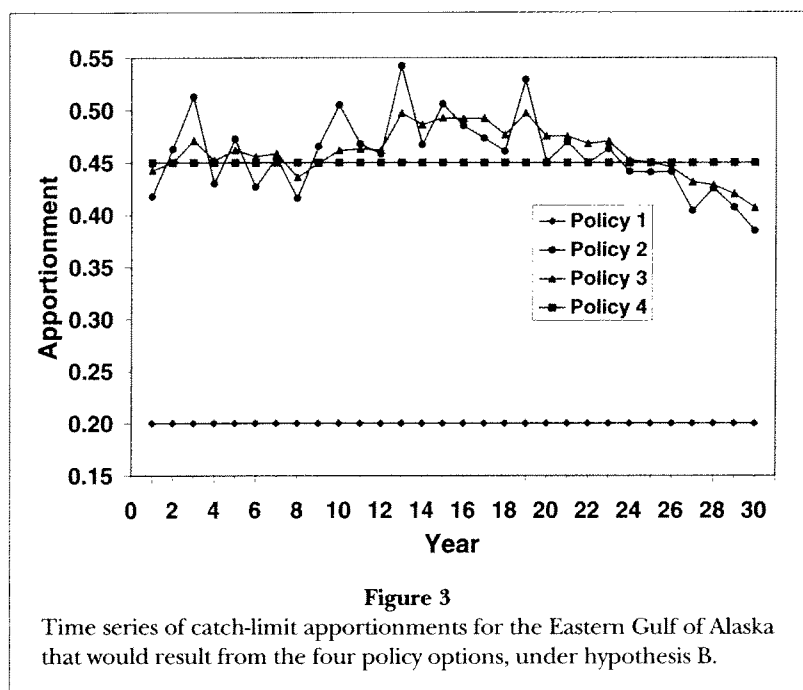
The unfished level of spawning biomass is found by letting $\mu = 0$.

It may be contended that the definition of risk chosen is unfounded because we have assumed that recruitment is independent of spawning biomass. We believe this risk is still a useful attribute for policy comparisons, because there is considerable uncertainty about the relationship between stock and recruitment for sablefish. We assume that CPUE is proportional to the density of fish in an area, where density is defined as an

² Sigler, M. Auke Bay Lab., NMFS, NOAA, 11305 Glacier Hwy., Juneau, AK 99801-8626. Personal commun., March 1993.

³ Fujioka, J.T. 1992. Sablefish. In Gulf of Alaska Groundfish Plan Team (eds.), Stock assessment and fishery evaluation report for the 1993 Gulf of Alaska groundfish fishery, section 4. N. Pac. Fish. Manage. Council, P.O. Box 103136, Anchorage, AK 99510-2252.

⁴ Lowe, S. A. 1992. Sablefish. In Bering Sea and Aleutian Islands Plan Team, (eds.), Stock assessment and fishery evaluation report for the groundfish resources of the Bering Sea/Aleutian Islands regions as projected for 1993, section 8. N. Pac. Fish. Manage. Council, P.O. Box 103136, Anchorage, AK 99510-2252.



area's available biomass divided by its available habitat for sablefish. Available habitat is defined as the combined area of gullies deeper than 400 m and continental shelf deeper than 200 m (Fujioka³). Attributes are presented for each regulatory area and for all regulatory areas combined.

Results

Geographic Yield per Recruit

Results of the geographic yield-per-recruit analysis are sensitive to assumptions about the geographic distribution of recruitment (Fig. 4). For hypothesis A (all areas have equal recruitment), the exploitation rate that results in maximum yield per recruit ($\mu_{\max}$) is much lower for the Eastern Gulf of Alaska regulatory area (EGOA) than for the other areas. As exploitation rate increases, a higher proportion of sablefish are caught in the more westward areas before they can return to the EGOA. Thus the highest yield for the EGOA is obtained under low fishing rates where fish are allowed to grow, mature, and move to the EGOA. For hypothesis B—most recruitment originates in the Western Gulf of Alaska (WGOA), Central Gulf of Alaska (CGOA), and EGOA—the same trend is evident. For hypothesis C (all recruitment originates in the EGOA), $\mu_{\max}$ is much lower for the Aleutians (AL) and to some extent the Eastern Bering Sea (EBS) and WGOA than for the more eastward areas. Under this hypothesis, the westward areas depend on migrants from the EGOA because no recruitment occurs

in the westward areas. With a low μ , fish that recruit to the eastward areas can move westward before they are caught.

Alternative Apportionment Policies

Results differ among the regulatory areas, but the three recruitment hypotheses provide similar results for most attributes (Table 5). Thus, results apply to the average of the three recruitment hypotheses.

For the easternmost regulatory areas (EGOA and CGOA), average annual yields are lowest for policy 1 (equal apportionment) and similar for the other three policies. Yields for policy 1 are only 43% of the yields for the other policies in EGOA and about 74% in CGOA. The pattern is reversed in the more westward areas (WGOA, EBS, AL), where policy 1 has the greatest yield. For these westward areas, yields for policies 2–4 were only about

20%–70% of the yield obtained for policy 1. For all areas combined, yields are lowest for policy 1 and similar among the other policies.

Regardless of area, yield varies most for policy 2. For the EGOA yield varies least for policy 3, whereas for the CGOA and the more westward areas, yield varies least for policy 4. For all areas combined, yield varies least for policy 1 and most for policy 2.

In the EGOA and (marginally) the CGOA, CPUE is greatest for policy 1 and similar for the other policies. In the more westward areas, CPUE for policies 2–4 is similar, but is lower for policy 1. For these westward areas, CPUE for policy 1 is 65%–80% of the maximum CPUE within an area. For all areas combined, CPUE is similar for policies 2–4 and substantially lower for policy 1.

In the EGOA, the risk of overfishing (the proportion of the time spawning biomass is below 30% of the unexploited level) is lowest for policy 1 and similar among the other policies. In the CGOA, all policies had the same level of risk. In the WGOA, BS, and AL, risk is greatest for policy 1. For policy 1 in these westward areas, spawning biomass is below 30% of the unexploited level 13%–48% of the time compared to only 3%–6% of the time for the other policies. For all areas combined, risk is slightly greater for policy 1 and identical among the other policies.

Discussion

Our goal was to incorporate movement patterns of sablefish into a yield-per-recruit analysis, evaluate alter-

native apportionment policies, and examine the sensitivity of results to different hypotheses about geographic distribution of recruitment. Including movement rates in an age-structured model of sablefish has increased our understanding of how the fisheries for sablefish in different areas may interact with each other.

The yield per recruit in an area is a function of exploitation rate, movement rate, and the distribution of recruitment. If the populations in the westward areas are mostly maintained by migrants from other areas and not by recruitment, as hypothesized by Sasaki (1985) and Beamish and McFarlane (1988), then fisheries in the westward areas may benefit from low overall exploitation rates. Conversely, if recruitment occurs throughout the northeastern Pacific Ocean, then fisheries in the eastern areas will benefit from low overall fishing rates. Future analyses should focus on estimating the geographical distribution of recruitment from age-composition data collected during surveys.

In our study, yield per recruit was evaluated under one apportionment strategy and with constant overall recruitment. A more comprehensive analysis of yield per recruit should include analysis of alternative policies and alternative stock-recruitment relationships.

In our results, we presented an evaluation of apportionment policies on the basis of the current estimated exploitation rate of 10%. We have also evaluated these policies with considerably higher exploitation rates (i.e., 20% and 30%), and the results were qualitatively similar.

Which is the most appropriate apportionment policy is unclear, because a particular policy did not dominate across all hypotheses and attributes. Policies 2–4 performed similarly across all attributes because, on average, the apportionments for these three policies are similar. Policy 1 had the lowest overall variability in yield but substantially higher risk in the westward areas, and the lowest overall CPUE. If an objective is to optimize CPUE and maintain spawning biomass on an areal basis, then policy 1 is not appropriate. Given the uncertainty about the location of sablefish spawning areas, a policy that maintains spawning biomass in each area is desirable.

Policies 2 and 3 have some appealing attributes. They adapt to current information about geographical distribution of available biomass and do not require estimates of movement rates for implementation. Policy 3 tends to produce less variable yields than policy 2 be-

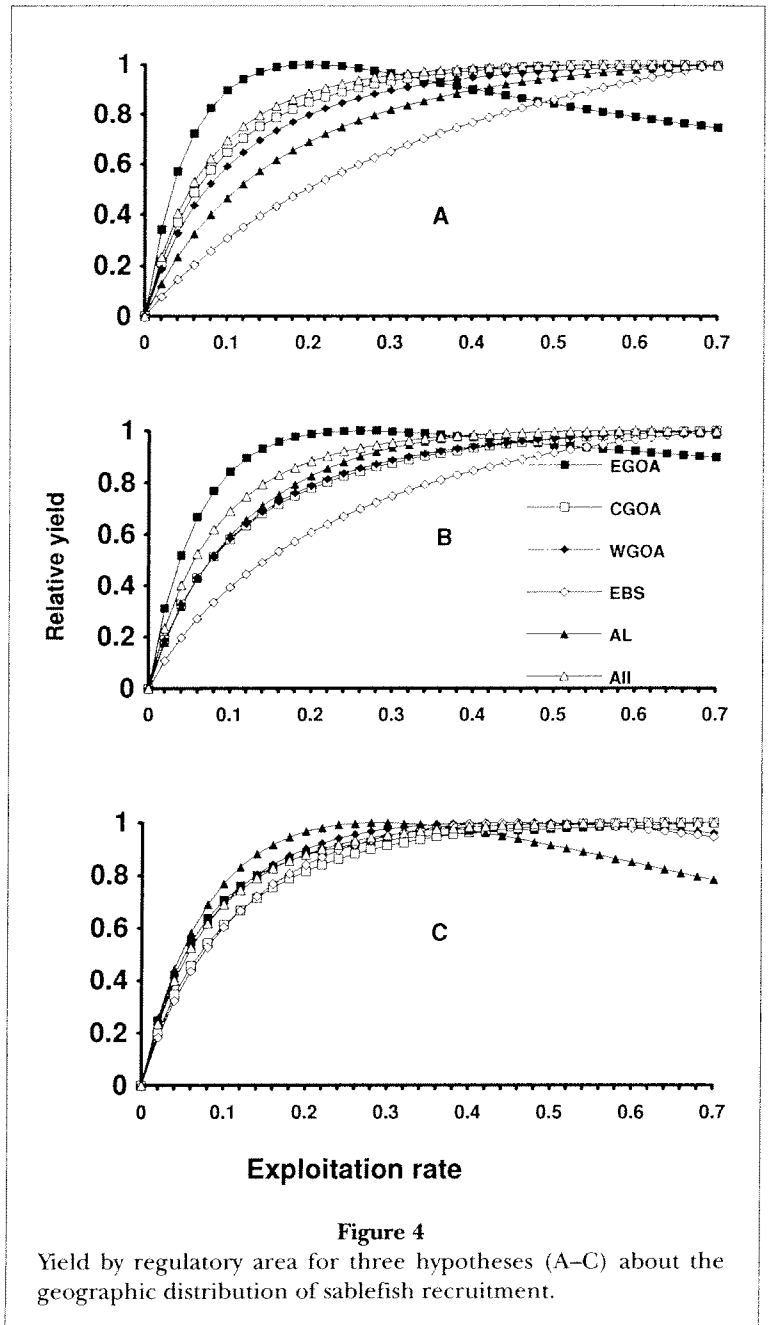


Figure 4
Yield by regulatory area for three hypotheses (A–C) about the geographic distribution of sablefish recruitment.

cause policy 3 is less sensitive than policy 2 to annual fluctuations in biomass estimates due to measurement error. For policy 3, the most recent year is given a weight of two and the prior four years are each given a weight of one. This weighting scheme has been chosen somewhat arbitrarily, and alternative weighting schemes may be more appropriate. If the method of weighting is based on the ratio of survey measurement error to total variability (i.e., measurement error plus variability in natural mortality, recruitment, and movement rates), then an exponential weighting scheme is appropriate (Meinhold and Singpurwalla, 1983).

Table 5

Average yield, coefficient of variation (CV) of yield, CPUE, and risk for four policies of apportionment of sablefish catch limits. Yield and CPUE are expressed relative to the maximum for a particular hypothesis.

		Hypothesis															
		A				B				C				Average			
Area ¹	Policy	Yield	CV	CPUE	Risk	Yield	CV	CPUE	Risk	Yield	CV	CPUE	Risk	Yield	CV	CPUE	Risk
EGOA	1	0.19	0.20	1.00	0.07	0.19	0.20	1.00	0.05	0.20	0.20	1.00	0.04	0.20	0.20	1.00	0.06
	2	0.42	0.21	0.94	0.10	0.46	0.21	0.93	0.09	0.52	0.21	0.90	0.08	0.46	0.21	0.92	0.09
	3	0.41	0.18	0.94	0.10	0.46	0.18	0.93	0.09	0.52	0.18	0.90	0.08	0.46	0.18	0.92	0.09
	4	0.41	0.19	0.94	0.10	0.45	0.19	0.93	0.08	0.51	0.19	0.90	0.08	0.46	0.19	0.92	0.09
CGOA	1	0.19	0.20	0.59	0.05	0.19	0.20	0.57	0.05	0.20	0.20	0.51	0.05	0.20	0.20	0.56	0.05
	2	0.25	0.22	0.60	0.05	0.27	0.22	0.57	0.05	0.27	0.22	0.49	0.06	0.27	0.22	0.55	0.05
	3	0.25	0.20	0.60	0.05	0.27	0.20	0.57	0.05	0.27	0.20	0.49	0.06	0.27	0.20	0.55	0.05
	4	0.26	0.19	0.60	0.05	0.27	0.19	0.57	0.05	0.27	0.19	0.49	0.06	0.27	0.19	0.55	0.05
WGOA	1	0.19	0.20	0.60	0.11	0.19	0.20	0.50	0.13	0.20	0.20	0.38	0.15	0.20	0.20	0.49	0.13
	2	0.15	0.22	0.69	0.04	0.15	0.22	0.59	0.04	0.13	0.22	0.45	0.06	0.14	0.22	0.56	0.05
	3	0.15	0.21	0.68	0.04	0.15	0.21	0.59	0.04	0.13	0.21	0.45	0.06	0.14	0.21	0.56	0.05
	4	0.15	0.19	0.68	0.04	0.15	0.19	0.59	0.05	0.13	0.19	0.45	0.06	0.14	0.19	0.56	0.05
EBS	1	0.19	0.20	0.07	0.14	0.19	0.20	0.03	0.29	0.20	0.20	0.02	0.46	0.20	0.20	0.04	0.30
	2	0.05	0.28	0.08	0.02	0.04	0.28	0.05	0.03	0.02	0.28	0.03	0.05	0.04	0.28	0.05	0.03
	3	0.05	0.27	0.08	0.02	0.04	0.27	0.05	0.03	0.02	0.27	0.03	0.06	0.04	0.27	0.05	0.04
	4	0.05	0.19	0.08	0.03	0.04	0.19	0.05	0.03	0.02	0.19	0.03	0.06	0.04	0.19	0.05	0.04
AL	1	0.19	0.20	0.09	0.14	0.19	0.20	0.04	0.41	0.20	0.20	0.02	0.89	0.20	0.20	0.05	0.48
	2	0.13	0.25	0.11	0.02	0.09	0.25	0.07	0.03	0.06	0.25	0.04	0.06	0.09	0.25	0.07	0.04
	3	0.13	0.24	0.11	0.02	0.09	0.24	0.07	0.03	0.06	0.24	0.04	0.06	0.09	0.24	0.07	0.04
	4	0.13	0.19	0.11	0.03	0.09	0.19	0.07	0.04	0.06	0.19	0.04	0.06	0.09	0.19	0.07	0.04
All areas combined	1	0.97	0.09	0.47	0.07	0.97	0.09	0.43	0.07	1.00	0.09	0.38	0.07	0.98	0.09	0.43	0.07
	2	1.00	0.11	0.67	0.05	1.00	0.12	0.67	0.06	1.00	0.13	0.66	0.07	1.00	0.12	0.66	0.06
	3	1.00	0.10	0.67	0.05	1.00	0.11	0.67	0.06	1.00	0.12	0.66	0.07	1.00	0.11	0.66	0.06
	4	1.00	0.10	0.66	0.06	1.00	0.11	0.67	0.06	1.00	0.12	0.66	0.07	1.00	0.11	0.66	0.06

¹ EGOA, Eastern Gulf of Alaska; CGOA, Central Gulf of Alaska; WGOA, Western Gulf of Alaska; EBS, Eastern Bering Sea; AL, Aleutians.

Policy 4 does not depend on biomass estimation on an areal basis for implementation, but does require knowledge of movement patterns. In our analysis, we assumed there was no error in information about movement patterns and that movement patterns change with age but not over time. There is considerable uncertainty about the estimates of movement rates used in this study (Heifetz and Fujioka, 1991). In addition, movement rates may exhibit some temporal variability. Temporal shifts in movement rates may be related to population abundance, environmental factors, or food abundance (Beverton and Holt, 1957; MacCall, 1990; Polacheck, 1990).

We do not examine the determination of an optimal policy of apportionment of catch limit among regulatory areas, which requires the definition of a function that incorporates the objectives of fishery management such as maximizing yield and minimizing yield variability

and risk (e.g., Quinn et al., 1990b). Geographic optimization of harvest can produce higher sustained yields than harvests without geographic restrictions (MacCall, 1990).

In conclusion, for populations that have high mixing rates among geographic areas, such as sablefish in the northeastern Pacific Ocean, an apportionment policy based on the geographic distribution of estimated biomass, such as policies 2 and 3, is reasonable. In general, such policies are robust to alternative recruitment hypotheses. For implementation, such policies require estimates of biomass but do not require estimates of movement rates. The current apportionment policy for sablefish (policy 3) also reduces the effects of annual fluctuations in biomass estimates due to measurement error.

Applying our techniques to other migratory populations may prove worthwhile. Modifications such as the

inclusion of a functional stock–recruitment relationship or variability in movement rates should be examined.

Acknowledgments

We thank J. Ianelli for assistance in designing the computer program used in this paper. The reviews of two anonymous referees improved the quality of this paper.

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